

Todo Management API

Module: Developing Software Systems II_25_SE2

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1. How to Run the Project

1. Open the project in Visual Studio
2. Run the project
3. API will be available at:
<http://localhost:3087>

Swagger:

<http://localhost:3087/swagger>

Docker:

`docker-compose up --build`

2. API Description

The API allows users to:

- Register and login
- Create, update and delete todo items
- Mark todos as completed or uncompleted
- View public todos without authentication
- Filter, sort and search todos
- Use pagination for listing todos

3. Database

The application uses a database with the following tables:

- Users
- TodoItems

4. Screenshots

Swagger UI

Auth			^
POST	/api/auth/register		🔒 ▼
POST	/api/auth/login		🔒 ▼
Todos			^
GET	/api/todos/public		🔒 ▼
GET	/api/todos		🔒 ▼
POST	/api/todos		🔒 ▼
GET	/api/todos/{id}		🔒 ▼
PUT	/api/todos/{id}		🔒 ▼
DELETE	/api/todos/{id}		🔒 ▼
PATCH	/api/todos/{id}/completion		🔒 ▼

Registration request

POST

/api/auth/register

🔒 ^

Parameters

Cancel

Reset

No parameters

Request body

application/json ▼

```
{  "email": "test@test.com",  "password": "123456",  "displayName": "Test"}
```

Execute

Clear

Responses

Curl

```
curl -X 'POST' \
  'http://localhost:3087/api/auth/register' \
  -H 'accept: text/plain' \
  -H 'Content-Type: application/json' \
  -d '{
    "email": "test@test.com",
    "password": "123456",
    "displayName": "Test"
  }'
```

Request URL

http://localhost:3087/api/auth/register

Server response

Code Details

409 Error: Conflict
Undocumented

Response body

```
{  "type": "https://httpstatuses.com/409",  "title": "Conflict",  "status": 409,  "detail": "A user with this email already exists."}
```

Response headers

```
content-type: application/problem+json; charset=utf-8
date: Tue, 31 Mar 2026 07:48:17 GMT
server: Kestrel
transfer-encoding: chunked
```

Responses		
Code	Description	Links
201	Created Media type text/plain Controls Accept header. Example Value Schema <pre>{ "id": "3fa85f64-5717-4562-b3fc-2c963f66fa6", "email": "string", "displayName": "string" }</pre>	No links

Login request

POST

/api/auth/login



Parameters

Cancel

Reset

No parameters

Request body

application/json

```
{  "email": "test@test.com",  "password": "123456"}
```

Execute

Clear

[illegible]

Responses		
Code	Description	Links
200	OK	No links
Media type		
text/plain		
Controls Accept header.		
Example Value Schema		
<pre>{ "accessToken": "string", "tokenType": "string", "expiresInSeconds": 0, "user": { "id": "3fa85f64-5717-4562-b3fc-2c963f66af60", "email": "string", "displayName": "string" } }</pre>		

Create todo

POST /api/todos

Parameters

No parameters

Request body

application/json

```
{
  "title": "Test task",
  "details": "Example task",
  "priority": "medium",
  "dueDate": "2026-04-10",
  "isPublic": true
}
```

Execute

Clear

Curl

```
curl -X 'POST' \
  'http://localhost:3087/api/todos' \
  -H 'accept: */*' \
  -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiI0MTQwNzI1IiwiaWF0IjoiMTQ1VWVOTDA2ZjZlMmR0RlNjM3ZDk0ZmV1LlCjlbW9pbC3RAd0VzdC5jb201L3JodHw0S1Rvc2NoZm1hcys4bWxzZ2FwLm9yZy83cy8yMDAxLlZl' \
  -H 'Content-Type: application/json' \
  -d '{
    "title": "Test task",
    "details": "Example task",
    "priority": "medium",
    "dueDate": "2026-04-10",
    "isPublic": true
  }'
```

Request URL

http://localhost:3087/api/todos

Server response

Code	Details
201	Response body

```
{
  "id": "b05ed399-332c-4b4b-8c19-56e00fe0df58",
  "title": "Test task",
  "details": "Example task",
  "priority": "medium",
  "dueDate": "2026-04-10",
  "isCompleted": false,
  "isPublic": true,
  "createdAt": "2026-03-31T07:59:03.4078549Z",
  "updatedAt": "2026-03-31T07:59:03.4078552Z"
}
```

Response headers

```
content-type: application/json; charset=utf-8
date: Tue, 31 Mar 2026 07:52:03 GMT
location: http://localhost:3087/api/todos/b05ed399-332c-4b4b-8c19-56e00fe0df58
server: Kestrel
transfer-encoding: chunked
```

Responses

Code	Description	Links
200	OK	No links

Get todos

GET

/api/todos

Parameters

Cancel

Name	Description
page integer(\$int32) (query)	1
pageSize integer(\$int32) (query)	10
status string (query)	all
priority string (query)	priority
dueFrom string(\$date) (query)	dueFrom
dueTo string(\$date) (query)	dueTo
sortBy string (query)	createdAt
sortDir string (query)	desc

Responses		
Curl<pre>curl -X 'GET' \ 'http://localhost:3007/api/todos?page=1&pageSize=10&status=all&sortBy=createdAt&sortDir=desc' \ -H 'accept: */*' \ -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIIOlI0MTAxNzBlZjQ4MjlhLTQ1YmVOTAAzMjZkZWY1ICJlbWVpbC16bnRlc3RadGVzdc5jb281CjodHRwOiRvc2NoZXhhcy44bXZlPwJmSyZy93cy8yMDA1LzA='</pre>		
Request URL<pre>http://localhost:3007/api/todos?page=1&pageSize=10&status=all&sortBy=createdAt&sortDir=desc</pre>		
Server response		
CodeDetails		
200Response body<pre>{ "items": [{ "id": "ba5d399-332c-4b4b-8c19-56e0f6df58", "title": "Test task", "details": "Example task", "priority": "medium", "dueDate": "2026-04-10", "isCompleted": false, "isPublic": true, "createdAt": "2026-03-31T07:59:03.407854Z", "updatedAt": "2026-03-31T07:59:03.407855Z" }, { "id": "d6c3ff4-c8bc-4a5a-9fd8-8ecb293f8f9", "title": "Beta Updated", "details": "Updated beta details", "priority": "high", "dueDate": "2026-04-11", "isCompleted": true, "isPublic": false, "createdAt": "2026-03-31T06:51:03.134836Z", "updatedAt": "2026-03-31T06:51:30.726462Z" }, { "id": "1136765c-b37c-4842-8d7e-418dc164f098", "title": "Alpha Updated", </pre>		
Response headers<pre>content-type: application/json; charset=utf-8 date: Tue, 31 Mar 2026 08:00:04 GMT server: Kestrel transfer-encoding: chunked</pre>		
Responses		
CodeDescriptionLinks		
200	OK	No links

Get public todos

GET

/api/todos/public

Parameters

Cancel

Name	Description
page integer(\$int32) (query)	1
pageSize integer(\$int32) (query)	10
status string (query)	all
priority string (query)	priority
dueFrom string(\$date) (query)	dueFrom
dueTo string(\$date) (query)	dueTo
sortBy string (query)	createdAt
sortDir string (query)	desc

```
Responses
```

```
curl -X 'GET' \
'http://localhost:3087/api/todos/public?page=1&pageSize=10&status=all&sortBy=createdAt&sortDir=desc' \
-H 'accept: */*' \
-H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpvcC9yLmFkbGkiOjEwMTAxNzBlZiM4dWQ1bnVlcnBpcC61nRlc3RADGVzdDcySjY2bWlyZWlnbm9rLnQ1YmVudAIIIA'
```

```
Request URL
```

```
http://localhost:3087/api/todos/public?page=1&pageSize=10&status=all&sortBy=createdAt&sortDir=desc
```

```
Server response
```

Code	Details
200	Response body <pre>{ "items": [{ "id": "b85ed399-332c-4b4b-8c19-56e0ef6ddf58", "title": "test task", "details": "Example task", "priority": "medium", "dueDate": "2026-04-10", "isCompleted": false, "isPublic": true, "createdAt": "2026-03-31T07:59:03.407854Z", "updatedAt": "2026-03-31T07:59:03.407855Z" }, { "id": "1136765c-b37c-4842-8d7e-418dc164f098", "title": "Alpha Updated", "details": "Updated alpha details", "priority": "medium", "dueDate": "2026-04-10", "isCompleted": true, "isPublic": true, "createdAt": "2026-03-31T06:50:23.19611Z", "updatedAt": "2026-03-31T06:51:26.980059Z" }, { "id": "8a50925b-4ffb-4655-956d-e7b48dd693fb", "title": "Alpha task", </pre>

Response headers		
<pre>content-type: application/json; charset=utf-8 date: Tue, 31 Mar 2026 08:03:46 GMT server: Kestrel transfer-encoding: chunked</pre>		
Responses		
Code	Description	Links
200	OK	No links

Completion update

PATCH

/api/todos/{id}/completion



Parameters

Cancel

Name	Description
id * required string(\$uuid) (path) b050d399-332c-4b4b-8c19-56e00f60df58	

Request body

application/json

```
{
  "isCompleted": true
}
```

Execute

Clear

Responses

Curl

```
curl -X "PATCH" \
"http://localhost:3007/api/todos/b05ed399-332c-4b4b-8c19-56e0f6edf58/completion" \
-H "accept: */*" \
-H "Authorization: bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiIiMTAxNzBlZjQwMjhlLHQlYmV0dA21ZWtWGRlbnHJ3Zk0zMVliCjE3bmFpbC16TnRlc3BAdGVrdC5jb281LCJodHRwOiJpc2NoZXMhcyS4bXNzb2FwLnBye3cy9y3cy8yMDAlIA==" \
-H "Content-Type: application/json" \
-d '{
  "isCompleted": true
}'
```

Request URL

```
http://localhost:3007/api/todos/b05ed399-332c-4b4b-8c19-56e0f6edf58/completion
```

Server response

CodeDetails

200

Response body

```
{
  "id": "b05ed399-332c-4b4b-8c19-56e0f6edf58",
  "title": "Test task",
  "details": "Example task",
  "priority": "medium",
  "dueDate": "2026-04-10",
  "isCompleted": true,
  "isPublic": true,
  "createdAt": "2026-03-31T07:59:03.407854Z",
  "updatedAt": "2026-03-31T08:06:03.6668208Z"
}
```

Download

Response headers

```
content-type: application/json; charset=utf-8
date: Tue, 31 Mar 2026 08:06:03 GMT
server: Kestrel
transfer-encoding: chunked
```

Responses

CodeDescriptionLinks

200OKNo links

Response headers		
<pre>content-type: application/json; charset=utf-8 date: Tue, 31 Mar 2026 08:08:13 GMT server: Kestrel transfer-encoding: chunked</pre>		
Responses		
Code	Description	Links
200	OK	No links

Database tables

Максимальное количество строк: 1000					
	Id	FullName	Email	PasswordHash	CreatedAt
▶	1	string	user@example...	\$2a\$11\$fCMEI7...	30.03.2026 19:...
⚙	NULL	NULL	NULL	NULL	NULL

TodoItems table (data created via API)

Curl

```
curl -X 'POST' \
  'http://localhost:3087/api/todos' \
  -H 'accept: */*' \
  -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiI0MTAxNzBlZi04MjI1LTQ1YmVtOTAzZjI1MGRlbnR3ZDk6ZWVlCjI1bmFpbC13RAd0VzdC5jb281LCJodHRwOi8vc2NoZmV1cy40bWZ2FwLm9yZy93cy8yMDA1LzA' \
  -H 'Content-Type: application/json' \
  -d '{
    "title": "Test todo",
    "description": "Test",
    "isCompleted": false,
    "priority": "Low"
  }'
```

Request URL

http://localhost:3087/api/todos

Server response

CodeDetails

201

Undocumented

Response body

```
{
  "id": "06a1feb7-b234-4984-9794-88d1d4f9f94c",
  "title": "Test todo",
  "details": null,
  "priority": "low",
  "dueDate": null,
  "isCompleted": false,
  "isPublic": false,
  "createdAt": "2026-03-31T09:31:59.6048264Z",
  "updatedAt": "2026-03-31T09:31:59.6048265Z"
}
```

Download

Response headers

```
content-type: application/json; charset=utf-8
date: Tue, 31 Mar 2026 09:31:58 GMT
location: http://localhost:3087/api/todos/06a1feb7-b234-4984-9794-88d1d4f9f94c
server: Kestrel
transfer-encoding: chunked
```

Responses

CodeDescriptionLinks

200

OK

No links

5. Sample Request and Response

GET /api/todos

Response:

```
{
  "items": [...],
```

```
"page": 1,  
"pageSize": 10,  
"totalItems": 2,  
"totalPages": 1  
}
```